# Thursday discussion

- Read paper
- Submit reflection before class

# Individual project

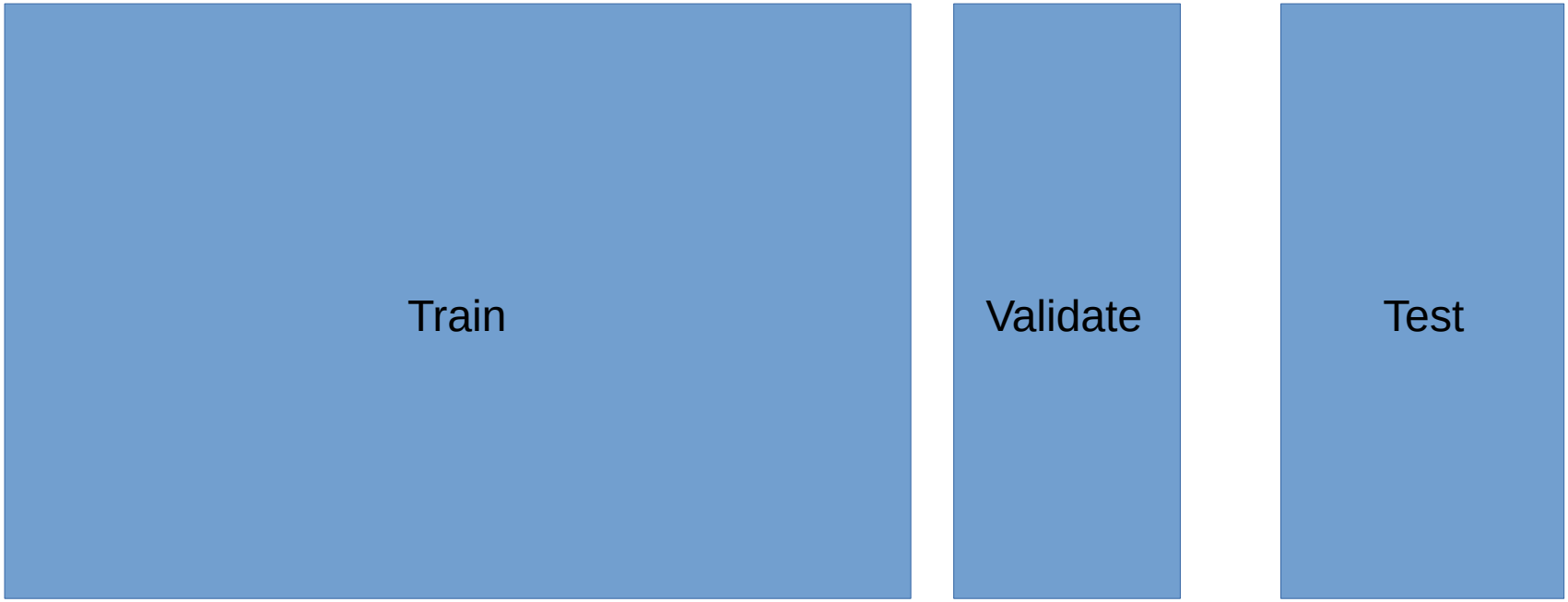
- Coding or literature review?
- Own data or other data?
  - Replications or improvements to existing papers are a great option!
- Group projects are possible
- Scope: 4 weeks
- Products: 15 min presentation (exam week), submit code or essay/review

# Past individual projects

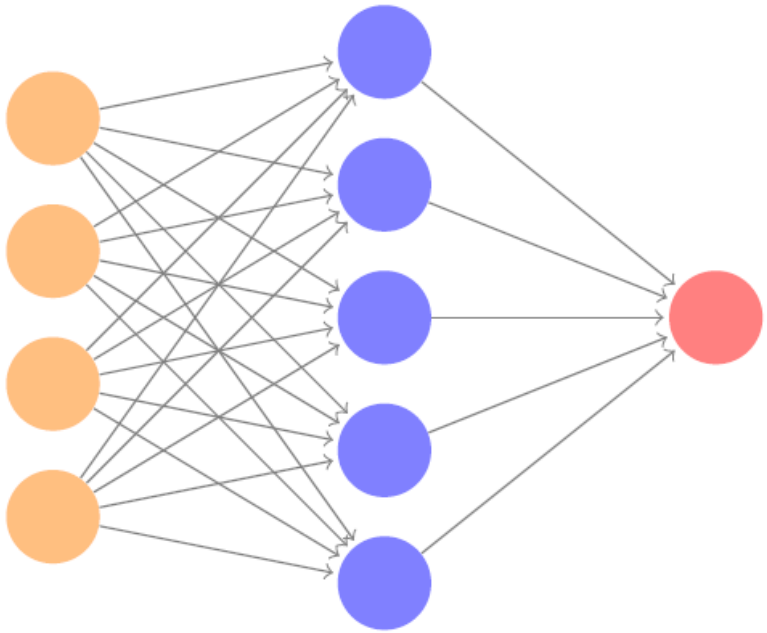
- Segmenting trees in drone imagery
- Mapping Pika habitat in aerial imagery
- Identifying fireflies by their flash pattern
- Predicting pH preference of microbial species
- Species distribution models of beetles
- What fish is this?
- Did this DNA come from a plasmid?

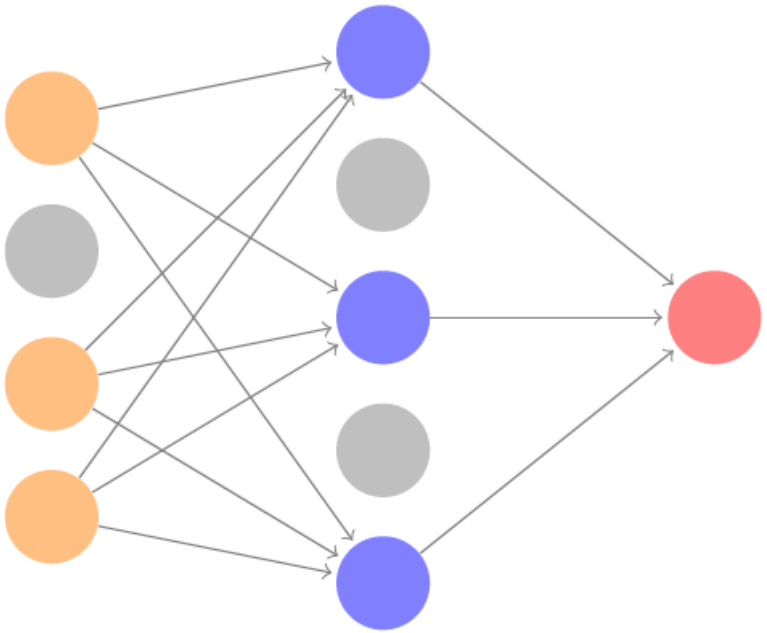
# Train-validate-test

- 3 way split, validate split is for tuning
- Test is a holdout to use after tuning/training



# Training algorithm: dropout





Like random forest sample columns

ISLR 10.19

# Today

- It's all about the data
  - How well can we generalize?
- Transfer learning & pre-trained models
  - code

# It's all about the data

- Generalization
- Out-of-distribution vs out-of-sample
- Scope of inference (scale, extent)
- Test set leakage

# Transfer learning

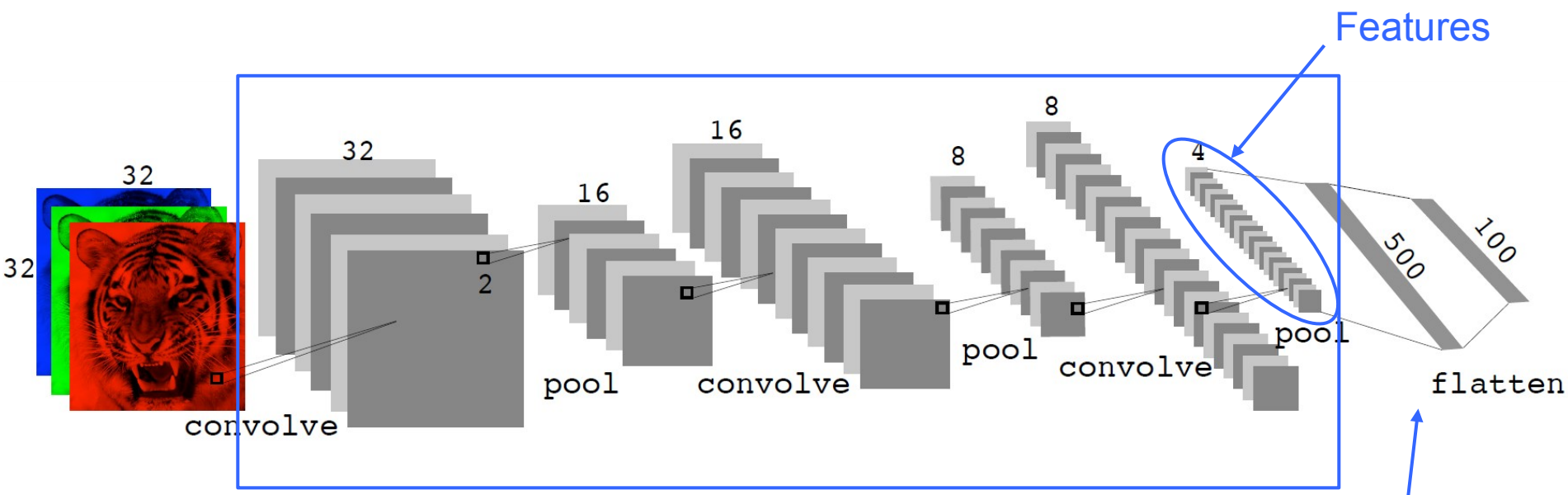
- Generalize some learned components from one task to another
- Kickstart the learning process on the target task

# Transfer learning

- Pretrained model
  - different dataset, related task
  - e.g. images from the internet, 1000 categories
  - vehicles, people, buildings, pets, nature, ...
- Fine tune or retrain existing or new layers
  - target data, target task
  - e.g. ecological taxa, 56 categories
  - plants, animals

# Transfer learning

Conv-pool learned to extract general features
Train final layers to classify images in new task



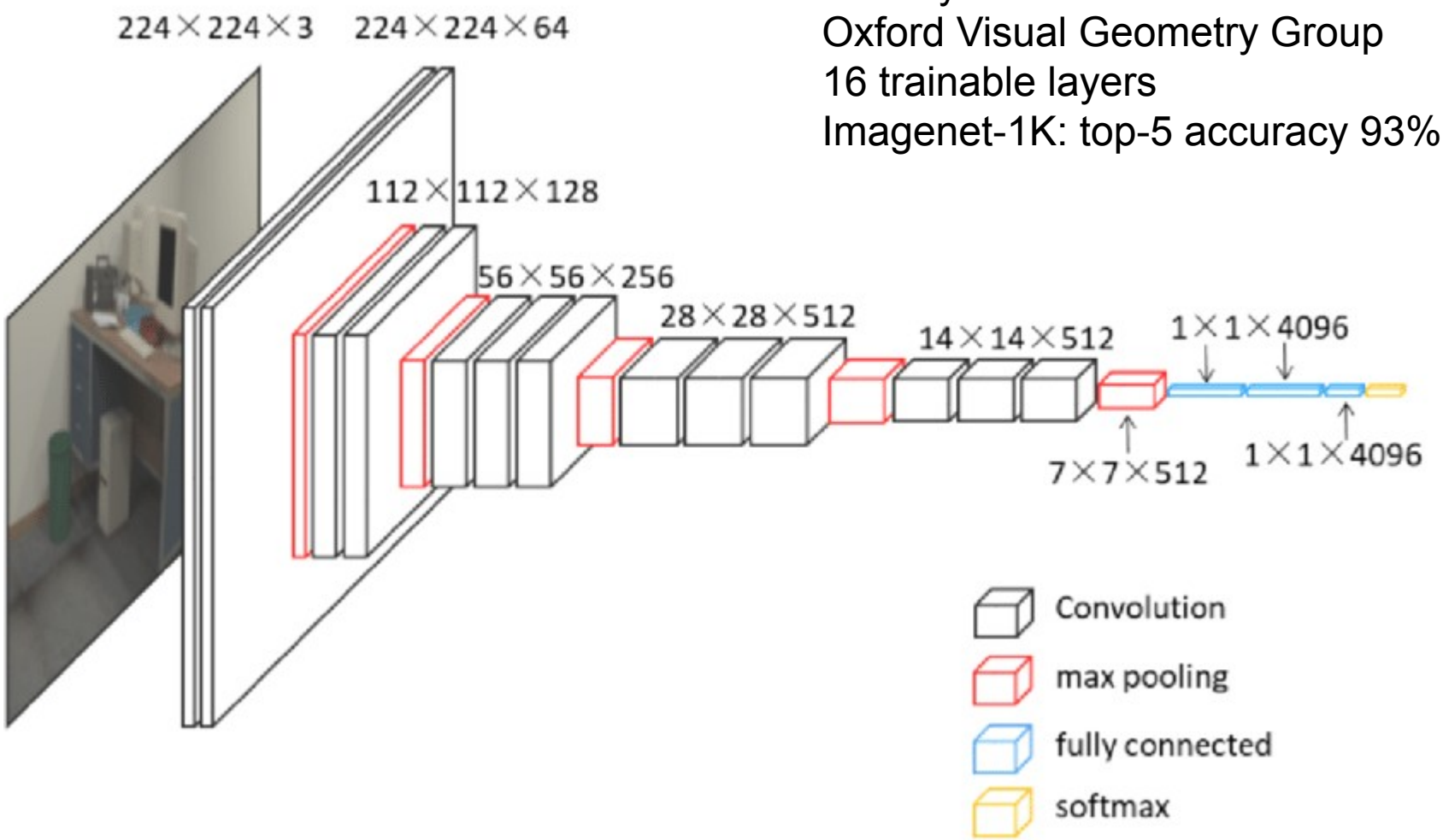

Freeze convolve-pool layers (architecture, weights etc)

Reconfigure and/or retrain dense layers

ISLR 10.8

# Imagenet challenge

- Imagenet: Deng et al (2009)
  - massive collection of labelled images
- Imagenet-1K
  - 1000 classes, mean ca 1200 images each
  - multi-resolution
- Alexnet: Krizhevsky, Sutskever, Hinton (2012)
  - CNN, GPU trained
  - top 5 accuracy improved from 70% to 85%
- Current top-5 accuracy: 99%

# VGG16

Simonyan & Zisserman 2014
Oxford Visual Geometry Group
16 trainable layers
Imagenet-1K: top-5 accuracy 93%